

CONFIGURATION FILE

```
veform-td-rec-0-2023_02_23-17_01_38_653.sigmf-meta  ue.sa.conf M  gnb.sa.band78.fr1.106PRB.usrpb210.conf M X
> PROJECTS > GENERIC-NR-5GC > CONF > gnb.sa.band78.fr1.106PRB.usrpb210.conf
{
    att_tx      = 12;
    att_rx      = 12;
    bands       = [78];
    max_pdschReferenceSignalPower = -27;
    max_rxgain   = 114;
    eNB_instances = [0];
    clock_src    = "internal";
}
);

rfsimulator :
{
    serveraddr = "server";
    serverport = 4043;
    options = ("saviq"); #("saviq"); or/and "chanmod"
    modelname = "AWGN";
    IQfile = "/tmp/rfsimulator.iqs";
};
```

- File used - [targets/PROJECTS/GENERIC-NR-5GC/CONF/gnb.sa.band78.fr1.106PRB.usrpb210.conf](#)
- Specifying “saviq” within the gNB conf file.

Generating samples by running gNB and UE.

```
ubuntu@ubuntu:~/openairinterface5g/cmake_targets/ran_build/build$ sudo ./nr-softmodem -O ../../targets/PROJECTS/GENERIC-NR-5GC/CONF/gnb.sa.band78.fr1.106PRB.usrpb210.conf --rfsim --gNBs.[0].min_rxtxtime 6

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
ubuntu@ubuntu:~/openairinterface5g/cmake_targets/ran_build/build$ sudo ./nr-uesoftmodem -r 106 --numerology 1 --band 78 -C 3619200000 --ssb 516 --rfsim
```

- `sudo ./nr-uesoftmodem -r 106 --numerology 1 --band 78 -C 3619200000 --ssb 516 --rfsim`

- `sudo ./nr-softmodem -O ../.././targets/PROJECTS/GENERIC-NR-5GC/CONF/gnb.sa.band78.fr1.106PRB.usrpb210.conf --gNBs.[0].min_rxtxttime 6 --rfsim`

Yes, these are IQ samples from the gNB, not UE”

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
[PHY] Got synch: hw slot offset 18, carrier off 7 Hz, rxgain 0.000000 (
DL 3619200007.000000 Hz, UL 3619200007.000000 Hz)
[PHY] UE synchronized! decoded_frame_rx=640 UE->init_sync_frame=0 trash
ed frames=300
[PHY] Resynchronizing RX by 276480 samples
[HW] received write reorder clear context
[NR_RRC] SIB1 decoded
[NR_MAC] TDD period index = 6, based on the sum of dl UL TransmissionPe
riodicity from Pattern1 (5.000000 ms) and Pattern2 (0.000000 ms): Total =
5.000000 ms
[NR_MAC] Set TDD configuration period to: 8 DL slots, 3 UL slots, 10 sl
ots per period (NR TDD_UL_DL_Pattern is 7 DL slots, 2 UL slots, 6 DL symb
ols, 4 UL symbols)
[NR_MAC] Configured 1 TDD patterns (total slots: pattern1 = 10, pattern
2 = 0)
[PHY] N_TA offset changed from 0 to 800
[MAC] Initialization of 4-Step CBRA procedure
[NR_MAC] PRACH scheduler: Selected R0 Frame 951, Slot 19, Symbol 8, Fdm
0
[PHY] PRACH [UE 0] in frame.slot 951.19, placing PRACH in position 2828
, Msg1/MsgA-Preamble frequency start 0 (kl 0), preamble_offset 2, first_n
onzero_root_idx 0, preambleIndex = 10
[PHY] RAR-Msg2 decoded
[NR_MAC] [UE 0][RAPROC][RA-RNTI 0113] Got BI RAR subPDU 5 ms
[NR_MAC] [UE 0][RAPROC][RA-RNTI 0113] Got RAPID RAR subPDU
[NR_MAC] [UE 0][RAPROC][952.7] Found RAR with the intended RAPID 10
[MAC] received TA command 31
[NR_MAC] [RAPROC][952.17] RA-Msg3 transmitted
[MAC] [UE 0][953.10][RAPROC] 4-Step RA procedure succeeded. CBRA: Conte
ntion Resolution is successful.
[NR_RRC] [UE0][RAPROC] Logical Channel DL-CCCH (SRB0), Received NR_RRCS
etup
[RLC] Added srb 1 to UE 0
[NR_RRC] State = NR_RRC_CONNECTED
[NAS] [UE 0] Received NR_NAS_CONN_ESTABLISH_IND: asCause 0
[MAC] [UE 0] Applying CellGroupConfig from gNodeB
[NAS] Generate Initial NAS Message: Registration Request
[NR_RRC] [UE 0][RAPROC] Logical Channel UL-DCCH (SRB1), Generating RRCS
etupComplete (bytes32)
[NR_PHY] [UE 0] RSRP = -42 dBm
[NR_MAC] Frame.Slot 768.0
[NR_MAC] Frame.Slot 896.0
[NR_PHY] [RAPROC] 951.19 Initiating RA procedure with preamble 10, ener
gy 43.0 dB (IO 0, thres 120), delay 0 start symbol 0 freq index 0
[NR_MAC] 951.19 UE RA-RNTI 0113 TC-RNTI 7d14: initiating RA procedure
[NR_MAC] UE 7d14: Msg3 scheduled at 952.17 (952.7 TDA 3) start 0 RBs 8
[NR_MAC] UE 7d14: 952.7 Generating RA-Msg2 DCI, RA RNTI 0x113, state 1,
preamble_index(RAPID) 10, timing offset = 0 (estimated distance 0.0 [m])
[NR_MAC] 952.7 Send RAR to RA-RNTI 0113
[NR_MAC] 952.17 PUSCH with TC RNTI 0x7d14 received correctly
[MAC] [RAPROC] Received SDU for CCCH length 6 for UE 7d14
[RLC] Activated srb0 for UE 32020
[RLC] Added srb 1 to UE 32020
[NR_MAC] Activating scheduling Msg4 for TC_RNTI 0x7d14 (state WAIT_Msg3
)
[NR_RRC] Decoding CCCH: RNTI 7d14, payload_size 6
[NR_RRC] [--] (cellID 0, UE ID 1 RNTI 7d14) Create UE context: CU UE ID
1 DU UE ID 32020 (rnti: 7d14, random ue id 2a7477c094000000)
[RLC] activate SRB 1 of UE 1
[NR_RRC] [DL] (cellID bc614e, UE ID 1 RNTI 7d14) Send RRC Setup
[NR_MAC] UE 7d14 Generate Msg4: feedback at 953.17, payload 225 bytes,
next state nrRA_WAIT_Msg4 MsgB ACK
[NR_MAC] 953.17 UE 7d14: Received Ack of Msg4. CBRA procedure succede
d (UE Connected)
[NR_MAC] Adding new UE context with RNTI 0x7d14
[NR_MAC] Frame.Slot 0.0
UE RNTI 7d14 CU-UE-ID 1 in-sync PH 0 dB PCMAX 0 dBm, average RSRP -44 (8
meas)
UE 7d14: dlslsch_rounds 1/0/0/0, dlslsch_errors 0, pucch0_DTX 0, BLER 0.10000
MCS (0) 0
UE 7d14: ulslsch_rounds 70/0/0/0, ulslsch_errors 0, ulslsch_DTX 0, BLER 0.05314
MCS (0) 0 (Qm 2 deltaMCS 0 dB) NPRB 5 SNR 50.5 dB
UE 7d14: MAC: TX 0 RX 210 bytes
UE 7d14: LCID 1: TX 0 RX 0 bytes
[NR_RRC] [UL] (cellID bc614e, UE ID 1 RNTI 7d14) Received RRCSetupCompl
ete (RRC_CONNECTED reached)
[NGAP] Selected PLMN in the NG Initial UE Message: MCC 1, MNC 1
[NGAP] No AMF is associated to the gNB

```

Verifying the stored samples.

```

oai_exit=1
• ubuntu@ubuntu:~/openairinterface5g/cmake_targets/ran_build/build$ hexdump
-C /tmp/rfsimulator.iqs | head
00000000  01 00 00 00 01 00 00 00  00 e8 b8 13 00 00 00 00  |.....
....|
00000010  00 00 00 00 00 00 00 00  f8 c0 7e 87 00 78 00 00  |.....~.
.X..|
00000020  01 00 00 00 00 e8 b8 13  00 00 00 00 00 00 00 00  |.....
....|
00000030  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  |.....
....|
*
0001e030  00 00 00 00 00 78 00 00  01 00 00 00 00 60 b9 13  |.....x.....
..|
0001e040  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00  |.....
....|
*
0003c040  00 00 00 00 00 00 00 00  00 00 00 00 00 78 00 00  |.....
.X..|
0003c050  01 00 00 00 00 d8 b9 13  00 00 00 00 00 00 00 00  |.....
....|
• ubuntu@ubuntu:~/openairinterface5g/cmake_targets/ran_build/build$ ls -lh
/tmp/rfsimulator.iqs
-rw-r--r-- 1 root root 5.2G Aug 18 09:46 /tmp/rfsimulator.iqs

```

- ls -lh /tmp/rfsimulator.iqs
- hexdump -C /tmp/rfsimulator.iqs | head

Replay_node and UE

```

^C
• ubuntu@ubuntu:~/openairinterface5g/cmake_targets/ran_build/build/radio/rfsimulator$ ./replay_node /tmp/rfsimulator.iqs 127.0.0.1 4043 DL

```

- ./replay_node /tmp/rfsimulator.iqs 127.0.0.1 4043 DL

```
ubuntu@ubuntu:~/openairinterface5g/cmake_targets/ran_build/build$ sudo ./nr-uesoftmodem -r 106 --numerology 1 --band 78 -C 3619200000  
--ssb 516 --rfsim --rfsimulator.serveraddr server --rfsimulator.serverport 4043
```

- `sudo ./nr-uesoftmodem -r 106 --numerology 1 --band 78 -C 3619200000 --ssb 516 --rfsim --rfsimulator.serveraddr server --rfsimulator.serverport 4043`

When these are run together, replay_node starts sending out samples to UE

```

sending at ts: 7376640, number of samples: 1920
sending at ts: 7378560, number of samples: 1920
sending at ts: 7380480, number of samples: 1920
sending at ts: 7382400, number of samples: 1920
sending at ts: 7384320, number of samples: 1920
sending at ts: 7386240, number of samples: 1920
sending at ts: 7388160, number of samples: 1920
sending at ts: 7390080, number of samples: 1920
sending at ts: 7392000, number of samples: 1920
sending at ts: 7393920, number of samples: 1920
sending at ts: 7395840, number of samples: 1920
sending at ts: 7397760, number of samples: 1920
sending at ts: 7399680, number of samples: 1920
sending at ts: 7401600, number of samples: 1920
sending at ts: 7403520, number of samples: 1920
sending at ts: 7405440, number of samples: 1920
sending at ts: 7407360, number of samples: 1920
sending at ts: 7409280, number of samples: 1920
sending at ts: 7411200, number of samples: 1920
sending at ts: 7413120, number of samples: 1920
sending at ts: 7415040, number of samples: 1920
sending at ts: 7416960, number of samples: 1920
sending at ts: 7418880, number of samples: 1920
sending at ts: 7420800, number of samples: 1920
sending at ts: 7422720, number of samples: 1920
sending at ts: 7424640, number of samples: 1920
sending at ts: 7426560, number of samples: 1920
sending at ts: 7428480, number of samples: 1920
sending at ts: 7430400, number of samples: 1920
sending at ts: 7432320, number of samples: 1920
sending at ts: 7434240, number of samples: 1920
sending at ts: 7436160, number of samples: 1920
sending at ts: 7438080, number of samples: 1920
sending at ts: 7440000, number of samples: 1920
sending at ts: 7441920, number of samples: 1920
sending at ts: 7443840, number of samples: 1920
sending at ts: 7445760, number of samples: 1920
sending at ts: 7447680, number of samples: 1920
sending at ts: 7449600, number of samples: 1920
sending at ts: 7451520, number of samples: 1920

[HW] Received data in past: current is 8110080, new reception:
7413120!
[HW] Received data in past: current is 8110080, new reception:
7415040!
[HW] Received data in past: current is 8110080, new reception:
7416960!
[HW] Received data in past: current is 8110080, new reception:
7418880!
[HW] Received data in past: current is 8110080, new reception:
7420800!
[HW] Received data in past: current is 8110080, new reception:
7422720!
[HW] Received data in past: current is 8110080, new reception:
7424640!
[HW] Received data in past: current is 8110080, new reception:
7426560!
[HW] Received data in past: current is 8110080, new reception:
7428480!
[HW] Received data in past: current is 8110080, new reception:
7430400!
[HW] Received data in past: current is 8110080, new reception:
7432320!
[HW] Received data in past: current is 8110080, new reception:
7434240!
[HW] Received data in past: current is 8110080, new reception:
7436160!
[HW] Received data in past: current is 8110080, new reception:
7438080!
[HW] Received data in past: current is 8110080, new reception:
7440000!
[HW] Received data in past: current is 8110080, new reception:
7441920!
[HW] Received data in past: current is 8110080, new reception:
7443840!
[HW] Received data in past: current is 8110080, new reception:
7445760!
[HW] Received data in past: current is 8110080, new reception:
7447680!
[HW] Received data in past: current is 8110080, new reception:
7449600!

```

After a while this happens, the synch fails

When using `replay_node` to stream these samples to the UE, the UE does start receiving them, but after some time the synchronization fails. I'm trying to understand why this desynchronization occurs even though the parameters match.

```
sending at ts: 257230080, number of samples: 1920
sending at ts: 257232000, number of samples: 1920
sending at ts: 257233920, number of samples: 1920
sending at ts: 257235840, number of samples: 1920
sending at ts: 257237760, number of samples: 1920
sending at ts: 257239680, number of samples: 1920
sending at ts: 257241600, number of samples: 1920
sending at ts: 257243520, number of samples: 1920
sending at ts: 257245440, number of samples: 1920
sending at ts: 257247360, number of samples: 1920
sending at ts: 257249280, number of samples: 1920
sending at ts: 257251200, number of samples: 1920
sending at ts: 257253120, number of samples: 1920
sending at ts: 257255040, number of samples: 1920
sending at ts: 257256960, number of samples: 1920
sending at ts: 257258880, number of samples: 1920
sending at ts: 257260800, number of samples: 1920
sending at ts: 257262720, number of samples: 1920
sending at ts: 257264640, number of samples: 1920
sending at ts: 257266560, number of samples: 1920
sending at ts: 257268480, number of samples: 1920
sending at ts: 257270400, number of samples: 1920
sending at ts: 257272320, number of samples: 1920
sending at ts: 257274240, number of samples: 1920
sending at ts: 257276160, number of samples: 1920
sending at ts: 257278080, number of samples: 1920
sending at ts: 257280000, number of samples: 1920
sending at ts: 257281920, number of samples: 1920
sending at ts: 257283840, number of samples: 1920
sending at ts: 257285760, number of samples: 1920
sending at ts: 257287680, number of samples: 1920
sending at ts: 257289600, number of samples: 1920
sending at ts: 257291520, number of samples: 1920
sending at ts: 257293440, number of samples: 1920
sending at ts: 257295360, number of samples: 1920
sending at ts: 257297280, number of samples: 1920
sending at ts: 257299200, number of samples: 1920
sending at ts: 257301120, number of samples: 1920
sending at ts: 257303040, number of samples: 1920
sending at ts: 257304960, number of samples: 1920
```

```
[PHY] SSB position provided
[NR PHY] Starting sync detection
[PHY] [UE thread Synch] Running Initial Synch
[NR PHY] Starting cell search with center freq: 3619200000, bandwidth: 106.
Scanning for 1 number of GSCN.
[NR PHY] Scanning GSCN: 0, with SSB offset: 516, SSB Freq: 0.000000
[PHY] synch Failed:
[PHY] SSB position provided
[NR PHY] Starting sync detection
[PHY] [UE thread Synch] Running Initial Synch
[NR PHY] Starting cell search with center freq: 3619200000, bandwidth: 106.
Scanning for 1 number of GSCN.
[NR PHY] Scanning GSCN: 0, with SSB offset: 516, SSB Freq: 0.000000
[PHY] synch Failed:
[PHY] SSB position provided
[NR PHY] Starting sync detection
[PHY] [UE thread Synch] Running Initial Synch
[NR PHY] Starting cell search with center freq: 3619200000, bandwidth: 106.
Scanning for 1 number of GSCN.
[NR PHY] Scanning GSCN: 0, with SSB offset: 516, SSB Freq: 0.000000
[PHY] synch Failed:
[PHY] SSB position provided
[NR PHY] Starting sync detection
[PHY] [UE thread Synch] Running Initial Synch
[NR PHY] Starting cell search with center freq: 3619200000, bandwidth: 106.
Scanning for 1 number of GSCN.
[NR PHY] Scanning GSCN: 0, with SSB offset: 516, SSB Freq: 0.000000
[PHY] synch Failed:
[PHY] SSB position provided
[NR PHY] Starting sync detection
[PHY] [UE thread Synch] Running Initial Synch
[NR PHY] Starting cell search with center freq: 3619200000, bandwidth: 106.
Scanning for 1 number of GSCN.
```